## Question #1 of 57

Question ID: 1675273

A Djiboutian (DJF) investor holds an international portfolio with beginning investments of USD 1,253,000 and EUR 2,347,800. Measured in the foreign currencies, these investments appreciate 5% and depreciate 7%, respectively.

Additional information:

| Beginning Spot Exchange Rate | Beginning Forward Exchange Rate | Ending Spot Exchange Rate |
|---|---|---|
| DJF/USD 179.54 | DJF/USD 185.67 | DJF/USD 192.85 |
| EUR/DJF 0.00416 | EUR/DJF 0.00413 | EUR/DJF 0.00421 |

The ending value of the EUR investment is *closest* to:

**A)** EUR 2,200,000.

**B)** DJF 575,000.

**C)** EUR 2,500,000.

**Explanation**

The ending value in EUR is: EUR 2,347,800 × 0.93 = EUR 2,183,454

The ending value in DJF is: EUR 2,183,454 / (EUR/DJF 0.00421) = DJF 518,635,154

(Module 18.1, LOS 18.a)

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## Question #2 of 57

Question ID: 1675293

A currency manager expects that the volatility of the JPY/USD (Japanese yen/U.S. dollar) currency pair will increase considerably. Of the set of appropriate strategies, the cheapest would *most likely* be the:

**A)** long straddle.

**B)** carry trade.

**C)** long strangle.

**Explanation**

Given that the manager expects an increase in volatility, a long option strategy is the most appropriate. A long strangle is cheaper than a long straddle because the strangle consists of buying an out-of-the-money call and put, whereas a long straddle consists of buying an at-the-money call and put.

A carry trade is appropriate in a stable market environment, but it could lead to significant uncertainty and losses if markets are volatile. Therefore, because of the potential losses, a carry trade is not likely to be the cheapest option.

(Module 18.4, LOS 18.e)

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## Question #3 - 6 of 57

Question ID: 1675304

In regard to the carry trade strategy, which of the following is *most accurate*?

**A)** Only Statement 1 is correct.

**B)** Both statements are correct.

**C)** Only Statement 2 is correct.

**Explanation**

*Statement 1 is correct*. The carry trade strategy is implemented by borrowing in a low-yielding currency and investing in a high-yielding currency. This strategy is profitable when the uncovered interest rate parity is violated. If the uncovered interest rate parity holds, the high-yielding currency would depreciate, on average, by the interest rate differential; therefore, the unhedged return on the high-yielding currency would match the return on the low-yielding currency and the return on the carry trade strategy would be zero.

*Statement 2 is also correct*. Historical data in FX markets shows that high-yielding currencies often appreciate rather than depreciate—a violation of the uncovered interest rate parity. Moreover, the data shows that the forward rate is not the center of the distribution of future FX spot rates, meaning that the forward rate is a biased estimate of future spot rates. Note that the carry trade is often referred to as trading the *forward rate bias*.

(Module 18.2, LOS 18.d)

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## Question #4 - 6 of 57

Question ID: 1675305

Based on Steffan's view and requirements, which of the following strategies should Bryan *most likely* recommend?

**A)** A 25-delta short risk reversal. ✗

**B)** A long straddle. ✗

**C)** A long 35-delta strangle. ✓

**Explanation**

A long straddle and a long strangle are both delta-neutral volatility trading strategies that attempt to profit from an increase in volatility.

A long straddle is implemented by buying an at-the-money call and at-the-money put, whereas a long strangle is implemented by buying an out-of-the-money call and out-of-the-money put. Because Steffan wants to minimize the cost of the strategy, a strangle is more appropriate than a straddle because the former incorporates out-of-the-money options, which are cheaper. So, a long 35-delta strangle is the correct answer.

A long straddle will profit in a case of an increase in volatility; however, given Steffan's requirement to minimize the cost of the strategy, the strangle is the most suitable strategy.

A 25-delta short risk reversal is implemented by buying a 25-delta put and selling a 25-delta call. This strategy is a directional trade rather than a volatility trade. The short risk reversal aims to provide downside protection below the strike of the 25-delta put and earn an income from selling the 25-delta call to partially (or fully) offset the cost of the put.

(Module 18.2, LOS 18.d)

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## Question #5 - 6 of 57

Question ID: 1675306

Steffan's reply pertaining to the roll yield on BRL forward contracts is *most likely*:

**A)** incorrect regarding the effect of the lower interest rate in Brazil. ✗

**B)** incorrect regarding the positive roll yield. ✓

**C)** correct. ✗

**Explanation**

The roll yield or roll return is the return from the movement of the forward price toward the spot price as the settlement of the forward contract approaches. The roll yield may be positive or negative, depending on whether the currency is bought or sold and whether it is trading at a forward premium or discount.

The BRL forward contract against the USD should trade at a discount because the interest rate in Brazil is higher than the interest rate in the United States. Selling BRL forward to hedge the currency exposure will generate a negative roll yield (because the clients are selling at a low price) rather than a positive roll yield (so, Steffan's reply is incorrect regarding the positive roll yield).

If Brazil had a lower interest rate when the forward contracts were initiated, the BRL would have been trading at a lower forward discount compared to the USD; thus, all else equal, selling the BRL forward would have generated a smaller negative roll yield, resulting in a lower cost of hedging.

(Module 18.4, LOS 18.f)

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## Question #6 - 6 of 57

Question ID: 1675307

Based on **Exhibit 1: Portfolio Allocation, Expected Standard Deviations, and Correlations**, the expected currency risk contribution to the risk of the domestic-currency returns of the portfolio is *closest* to:

**A)** 13.7%.

**B)** 9.3%.

**C)** 4.4%.

**Explanation**

The currency risk contribution is computed as the difference between the risk of the domestic-currency (DC) returns and the risk of the foreign-currency (FC) returns.

First, compute the variance of the FC returns of the portfolio as:

$$\sigma^2(R_{FC}) \approx w_1^2\sigma^2(R_1) + w_2^2\sigma^2(R_2) + 2w_1w_2\sigma(R_1)\sigma(R_2)\rho(R_1, R_2)$$

Where $R_{FC}$ is the FC return of the European portfolio (denominated in EUR); $R_1$ and $R_2$ are the FC returns of the European equities and bonds, respectively; $w_1$ and $w_2$ are the weights of equities and bonds in the portfolio, respectively; σ represents the standard deviation; and $\rho(R_1,R_2)$is the correlation of returns between European equities and bonds.

$\sigma^2(R_{FC}) \approx (0.6^2 \times 0.1444^2) + (0.4^2 \times 0.0426^2) + (2 \times 0.6 \times 0.4 \times 0.1444 \times 0.0426 \times 0.30) \approx 0.00868$; thus, the risk of the portfolio's FC returns = $\sigma(R_{FC}) \approx \sqrt{0.00868} = 9.32\%$.

Next, compute the variance of the DC returns of the portfolio as:

$$\sigma^2(R_{DC}) \approx \sigma^2(R_{FC}) + \sigma^2(R_{FX}) + 2\sigma(R_{FC})\sigma(R_{FX})\rho(R_{FC}, R_{FX})$$

where $R_{DC}$ is the DC return of the European portfolio (denominated in USD) and $R_{FX}$ is the return of the FC against the DC (return of the USD/EUR).

$\sigma^2(R_{DC}) \approx 0.0932^2 + 0.068^2 + (2 \times 0.0932 \times 0.068 \times 0.43) = 0.01876$; thus, the risk of the portfolio's DC returns = $\sigma(R_{DC}) \approx \sqrt{0.01876} \approx 13.70\%$.

Finally, the expected currency risk contribution =13.70% - 9.32% = 4.38%, so this is the correct answer.

The 9.3% answer option is incorrect as $R_{FC}$ rather than the currency risk contribution to the portfolio risk is calculated.

The 13.7% answer option is incorrect as $\sigma(R_{DC})$ rather than the currency risk contribution to the portfolio risk is calculated.

(Module 18.1, LOS 18.a)

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## Question #7 of 57

Question ID: 1675292

A currency trader wants to implement a long strangle strategy on a currency pair and is considering either a 10-delta or 40-delta strategy. Relative to the 10-delta strangle, the 40-delta strangle would:

**A)** be cheaper.

**B)** have the same intrinsic value.

**C)** have a more moderate payoff structure.

**Explanation**

A long strangle consists of a long position in both an out-of-the-money (OTM) call and put option with the same absolute deltas, maturities, and underlying currency. Because both options are OTM with the absolute value of their deltas less than 0.5, the intrinsic value of both options is zero.

The 40-delta strangle is less OTM (or more in the money) than the 10-delta strangle and would, therefore, be more expensive with a better payoff structure.

(Module 18.4, LOS 18.e)

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## Question #8 of 57

Question ID: 1675329

Which of the following statements regarding hedging of emerging-market currencies is *least accurate*?

**A)** Hedging costs for managers who buy emerging-market currencies are increased by the relatively high interest rates in emerging markets.

**B)** Hedging cost varies with normally large bid/ask spreads followed by infrequent periods of even higher spreads.

**C)** Tail risk and contagion both refer to relatively infrequent events that increase the difficulty of hedging emerging-market currencies.

**Explanation**

The relatively high interest rates of emerging-market economies leads to an inverted pricing curve with forward prices of the emerging-market currencies below their spot prices. This raises hedging costs for sellers of the currency, not buyers; sellers receive negative roll yield, while buyers receive positive roll yield. EM currencies do have relatively high bid/ask spreads, which increase in periods of crisis. Contagion and tail risk refer to infrequent events. Contagion refers to all EM currencies tending to decline together in periods of crisis, and tail risk to the downside in those periods of crisis being large in relation to typical upside movement in the currencies.

(Module 18.6, LOS 18.i)

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## Question #9 - 12 of 57

Question ID: 1675318

In regard to emerging market currencies, which of the following is *most accurate*?

**A)** Both notes are correct. ✗

**B)** Only Note 1 is correct. ✓

**C)** Only Note 2 is correct. ✗

**Explanation**

*Note 1 is correct*. The roll yield is the return from the convergence of the forward price toward the spot price as the forward contract nears expiry. Emerging market currencies are typically high-yield currencies due to the higher risk of emerging market economies, whereas developed market currencies are typically low-yield currencies. The higher-yielding currency (the emerging market currency) will trade at a forward discount against the lower-yielding currency; as such, hedging the long emerging market currency exposures involves selling the currency forward at a price that is lower than the current spot price, thereby creating a negative roll yield.

*Note 2 is incorrect*. Currency exposures may be hedged using forward or futures contracts; however, forward contracts are typically used due to several reasons, including the unavailability of the desired futures contracts or the illiquidity of emerging market currency futures contracts. Because forward contracts do not require posting and maintaining margins, margins do not have to be monitored.

(Module 18.4, LOS 18.f)

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## Question #10 - 12 of 57

Question ID: 1675319

Based on the information pertaining to KJ Endowment's U.K. assets and **Exhibit 1: USD/GBP Current Spot and Forward Rates**, to rebalance the USD/GBP hedge, Grasizi should *most likely*:

**A)** sell GBP forward at 1.3992. ✗

**B)** sell GBP forward at 1.3988. ✓

**C)** buy GBP forward at 1.4001. ✗

**Explanation**

To hedge the long GBP exposure, Grasizi sold a GBP 50 million, 6-month forward contract (i.e., the USD forward contract was bought). After three months, because the value of the U.K. equities has increased by GBP 3.5 million—while the value of U.K. bonds has decreased by GBP 0.8 million—the total value of the U.K. assets has increased by GBP 2.7 million; as such, Grasizi should sell an additional GBP 2.7 million forward for three months because he was initially requested to hedge the exposure for six months. The GBP would be sold using a 3-month forward contract at the bid price, which is equal to the spot bid price plus (or minus) the 3-month forward points. Because the spot price is quoted to four decimals, the forward points are divided by 10,000.

As such, the 3-month forward bid price = spot bid price + 3-month forward points = 1.3984 + 4 / 10,000 = 1.3988, so this is the correct answer.

The 6-month forward bid price is 1.3992. Adjusting the hedge requires selling GBP forward for three months because Grasizi was initially asked to hedge for six months and three months have passed.

Because the fund's GBP exposure increased, rebalancing the hedge requires selling an additional GBP 2.7 million, 3-month forward contract.

(Module 18.4, LOS 18.f)

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## Question #11 - 12 of 57

Question ID: 1675320

Based on the information collected about the University of Hubber Endowment, Grasizi should *most likely* recommend:

**A)** managing currency exposures with narrow tolerance bands.

**B)** a passive hedging strategy.

**C)** an active currency management strategy.

**Explanation**

Currency management strategies vary from passive hedging to treating the currency as a separate asset class to generate currency alpha. The currency management strategy should be included in the client's investment policy statement (IPS).

The primary objective of an active currency management strategy is to manage currency risks to add alpha to the portfolio.

A portfolio's currency management strategy is tilted toward active currency management when

- the portfolio has long-term investment objectives,
- the portfolio has a high risk tolerance,
- the portfolio has low liquidity needs,
- the portfolio has a high allocation to equities in foreign currencies, and
- the client believes in the benefits of active currency management.

Based on the information collected by Grasizi, the endowment has long-term investment objectives (perpetuity), high risk tolerance, low liquidity needs (given the large expected donations), and a large allocation to equities (including international equities). Moreover, the endowment's investment committee believes in Global Prime's ability to exploit inefficiencies in the foreign exchange market. As such, active currency management is the most appropriate strategy.

With a passive hedging strategy, the portfolio's currency exposures are matched with (or kept close to) those of the benchmark, and the goal is to eliminate currency risk relative to the benchmark. Based on the information collected by Grasizi, the characteristics of the endowment do not support a passive hedging strategy.

With a discretionary hedging strategy, managers are given limited discretion on how far they can deviate from the neutral position (i.e., the benchmark's currency exposures). The primary objective of this strategy is to reduce the portfolio's currency risk, while giving managers the ability to have some directional opinion to enhance portfolio returns. The characteristics of the endowment do not support a discretionary hedging strategy.

(Module 18.1, LOS 18.c)

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## Question #12 - 12 of 57

Question ID: 1675321

Given Soratis's objectives, which of the following is the *most appropriate* strategy for hedging the portfolio's USD exposure using options on USD/EUR?

**A)** Buy a 50-delta call and sell a 20-delta call. ✗

**B)** Buy a knock-in call. ✓

**C)** Buy a 20-delta put. ✗

**Explanation**

Because the options are based on USD/EUR, the calls and puts on USD/EUR refer to calls and puts on the base currency—the EUR. Because Soratis has long positions in U.S. equities, he is long the USD. To hedge the long USD exposure, Soratis should buy puts on the USD (i.e., calls on the EUR).

Buying a *knock-in call* on the USD/EUR means buying a call on the EUR or buying a put on the USD. Buying puts on the USD protects the position against the depreciation of the USD and maintains the upside potential. Because knock-in options are created only when the exchange rate touches a preset barrier, they are cheaper than plain vanilla options; therefore, using such options reduces the cost of the strategy. Of the three choices, buying a knock-in call is the only strategy that meets Soratis's objectives.

Buying a 20-delta put (i.e., an out-of-the-money put) on the EUR is equivalent to buying a 20-delta call (i.e., an out-of-the-money call) on the USD. Buying calls on the USD will increase the exposure to the USD, rather than hedge it.

Buying a 50-delta call (i.e., an at-the-money call) on the EUR and selling a 20-delta call (i.e., an out-of-the-money call) on the EUR is equivalent to buying a 50-delta put (i.e., an at-the-money put) on the USD and selling a 20-delta put (i.e., an out-of-the-money put) on the USD. Because the 50-delta put protects against the downside risk until the USD/EUR reaches the strike price of the 20-delta put (below which the currency exposure is not hedged), this strategy will not meet Soratis's objectives.

(Module 18.5, LOS 18.g)

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## Question #13 of 57

Question ID: 1675274

A European investor holds a diversified portfolio. From the euro perspective, the portfolio is weighted 60% and 40% in U.S. and U.K. investments.

Additional information:

| Assets: | Returns measured in foreign currency: | Returns measured from investor's perspective: | Standard deviation of asset's returns measured in foreign currency: | Stand deviation of the foreign currency's returns: |
|---|---|---|---|---|
| U.S. | 5% | 6% | 4.5% | 3.7% |
| U.K. | 7% | 8% | 3.5% | 4.7% |

The correlation between the foreign-currency asset's returns and returns on the foreign currency are 0.81 and 0.67, respectively, for the U.S. and U.K. assets. Which of the following is the standard deviation of returns for the investor in the U.K. assets?

**A)** 7.5%.

**B)** 7.8%.

**C)** 60.9%.

**Explanation**

It depends on the standard deviation of the asset returns measured in the foreign currency, the standard deviation of the currency's returns, and the correlation between these two sources of returns:

$$\text{Variance} = (1.0^2)(3.5^2) + (1.0^2)(4.7^2) + 2(1.0)(1.0)(0.67)(3.5)(4.7) = 56.38$$

Standard deviation = 7.5%

(Module 18.1, LOS 18.a)

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## Question #14 of 57

Question ID: 1675296

A trader has a EUR 5,000,000 short asset exposure and wishes to hedge it using a six-month long forward contract. Relative to choosing a one-month long forward contract that is rolled over each month, the six-month forward contract would *most likely*:

**A)** indicate lower risk aversion.

**B)** be a perfect hedge of the initial notional value.

**C)** have some realized gains or losses before forward maturity.

**Explanation**

The six-month forward contract requires no rebalancing, while the one-month rolled forward requires monthly rebalancing. Less frequent rebalancing indicates lower risk aversion.

Because the value of the hedged asset could change over the six months, the six-month hedge is not perfect and could result in a significantly overhedged or underhedged position (e.g., if the asset value increases to EUR 5,500,000 in two months, the manager would be underhedged by EUR 500,000).

Because there are no contracts to roll over before the six-month maturity, there are no realized gains or losses generated before the maturity of the six-month contract.

(Module 18.4, LOS 18.f)

## Question #15 of 57

Question ID: 1675327

A Canadian manager has a large portfolio with exposures to numerous foreign currencies, including the U.S. dollar (USD), the euro (EUR), the British pound (GBP), the Swiss franc (CHF), and some emerging-market currencies. When hedging the portfolio's foreign currency risk, which of following hedges would the manager *most likely* use?

**A)** Macro hedge. ✓

**B)** Direct hedge. ✗

**C)** Cross hedge. ✗

**Explanation**

A macro hedge is a type of cross hedge that addresses portfolio-wide risk factors rather than the risk of individual portfolio assets. One type of currency macro hedge uses a derivatives contract based on a fixed basket of currencies to modify currency exposure at a macro (portfolio) level. The currency basket in the contract may not precisely match the currency exposures of the portfolio, but it can be less costly than hedging each currency exposure individually. The manager must make a choice between accepting higher residual currency risk versus lower cost.

A direct hedge is possible for widely traded currencies such as the USD, EUR, and GBP. However, because the portfolio contains some emerging-market currencies, those currencies are not likely to be efficiently hedged using direct hedges due to high transaction costs and the potential nonexistence of an appropriate hedging contract.

A cross hedge is likely most efficient for the emerging-market currencies (to reduce transaction costs), but it is not necessary for widely traded currencies such as the USD, EUR, and GBP.

(Module 18.6, LOS 18.h)

## Question #16 of 57

Question ID: 1675291

Which scenario would *most likely* lead a portfolio manager to buy a foreign currency in the forward market?

**A)** An active currency manager who is underweighted in assets denominated in that currency. ✓

**B)** A portfolio that is overweighted in assets denominated in that currency. ✗

**C)** An active manager implementing a carry trade when the currency is being bought as the higher relative yield. ✗

**Explanation**

No manager views were indicated in the question; lacking views, an active manager should take a neutral exposure to the currency. Being underweighted in the assets would produce an underweighting in the currency, which is corrected by buying the currency forward. The other two answers are incorrect. Being overweighted would lead to selling the currency forward. A carry trade would buy the higher yield currency in the spot, not forward, market.

(Module 18.4, LOS 18.e)

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## Question #17 of 57

Question ID: 1675297

A Swiss-based investor (home currency is CHF) has a short exposure to the pound (GBP). The one-year CHF interest rate is 1.5% and the GBP interest rate is 2.5%. Using a one-year forward contract, the investor will *most likely*:

**A)** have a positive roll yield because the GBP is trading at a forward discount. ✓

**B)** have a negative roll yield because the GBP is trading at a forward discount. ✗

**C)** have a positive roll yield because the GBP is trading at a forward premium. ✗

**Explanation**

To hedge the short GBP exposure, the investor will buy the forward contract. Using the GBP as the base currency, based on interest rate parity, the GBP will trade at a forward discount because of the higher interest rate.

Forward = spot × $[(1 + i_{CHF}) / (1 + i_{GBP})]$ = spot × [(1.015) / (1.025)] = spot × 0.99

Therefore, the GBP forward discount is 0.99 – 1 = –0.01 or –1%. Because the forward is "cheaper" than the spot, the long exposure will earn a positive roll yield (think sell high, buy low).

(Module 18.4, LOS 18.f)

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## Question #18 of 57

Question ID: 1675288

Currency trading based on economic fundamentals would be *most likely* to sell a currency forward if the country issuing the currency is experiencing:

**A)** increasing real rates of return.

**B)** declining levels of relative risk in the economy.

**C)** rising relative inflation.

**Explanation**

Higher relative inflation is associated with a declining value of the currency, and it would tend to encourage a sale of the currency by the manager. The other two factors are associated with currency appreciation.

(Module 18.2, LOS 18.d)

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## Question #19 of 57

Question ID: 1675309

Cristina has a long exposure to the Brazilian real (BRL) and wants to use options to hedge against a modest depreciation of the BRL. She is primarily concerned with keeping upside potential as well as minimizing the cost of buying options. Which of the following derivative positions will *best* achieve Cristina's objectives?

**A)** Collar.

**B)** Put spread.

**C)** Seagull spread.

**Explanation**

Cristina can hedge a potential BRL depreciation with put options—although taking a long put option, especially if it is in the money, is expensive. Therefore, she can use a *put spread* to subsidize a portion of the cost of the long put. A put spread involves buying an out-of-the-money (OTM) put option and selling an even further OTM put option (e.g., the investor can buy a 40-delta BRL put and sell a 20-delta BRL put). The more OTM put will subsidize a portion of the long put's cost, although the investor loses downside protection below the more OTM put's strike price. If the BRL appreciates, the put options will be worthless, and the investor keeps all the upside potential.

A *collar* involves buying a put and selling a call with the same delta. The sale of the call subsidizes the cost of the long put. However, while the position provides downside protection, the short call position will limit the currency's upside potential.

A *seagull spread* essentially combines a put spread with a collar (buy an OTM put, sell a more OTM put, sell an OTM call); therefore, it achieves the objective of minimizing option costs but provides limited upside potential due to the short call position.

(Module 18.5, LOS 18.g)

## Question #20 of 57

Question ID: 1675285

Which of the following portfolios would *most likely* follow a passive currency hedging strategy?

**A)** One with more confidence in the portfolio manager and high income needs. ✗

**B)** One very concerned with minimizing regret and higher allocation to equity investments. ✗

**C)** One with a shorter time horizon and higher liquidity needs. ✓

**Explanation**

The following will shift the portfolio toward more passive currency management:

- A short time horizon for portfolio objectives
- High risk aversion
- Lack of concern with regret at missing opportunities to add value through discretionary currency management
- High short-term income and liquidity needs
- Significant foreign currency bond exposure
- Low hedging costs
- Clients who doubt the benefits of discretionary management

(Module 18.1, LOS 18.c)

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## Question #21 of 57

Question ID: 1675331

Which of the following statements regarding nondeliverable forwards (NDFs) is correct?

**A)** The pricing of NDFs reflects supply and demand conditions. ✓

**B)** The controlled currency is usually a major currency. ✗

**C)** NDFs usually have higher credit risk than regular forward contracts. ✗

**Explanation**

The pricing of NDFs does not necessarily follow covered interest rate parity; pricing reflects the supply and demand conditions in the offshore market, which may be different than the onshore market of the specific emerging-market country.

The *noncontrolled* currency is usually the USD or some other major currency. The controlled currency is the emerging-market currency.

The credit risk of an NDF is usually less than that of a regular forward contract because there is no delivery of the notional amounts required.

(Module 18.6, LOS 18.i)

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## Question #22 - 25 of 57

Question ID: 1675333

Compared to discretionary hedging, active currency management *most likely*:

**A)** focuses on reducing portfolio risk.

**B)** allows the manager less discretion in deviating from the benchmark.

**C)** focuses on increasing portfolio return.

**Explanation**

Active currency management most likely focuses on increasing portfolio return.

In active currency management, the goal is to create incremental return (alpha), not to reduce risk.

Active currency management allows a manager to have greater deviations from benchmark currency exposures, whereas discretionary hedging allows only modest deviations by a specified percentage. As a result, the manager is expected to generate positive incremental portfolio return from actively managing a portfolio's currency exposure.

(Module 18.1, LOS 18.b)

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## Question #23 - 25 of 57

Question ID: 1675334

Considering his portfolio's mission, Verse's strategy is *most appropriate* in the country of:

**A)** Panacea.

**B)** Madison.

**C)** Quincy.

**Explanation**

The country of Panacea is correct because Verse has been charged with using technical analysis in the currency markets. In Panacea, Verse is using the currency's momentum and moving averages. Moving averages of price are often used in technical analysis.

In Quincy and Madison, Verse is using interest rates and inflation to form a position. Interest rates and inflation are economic fundamentals. Technical analysis is based partly on the principle that it is unnecessary to know what the currency should be worth based on fundamental value. Therefore, Verse's strategy in Quincy and Madison is not consistent with the use of technical analysis, so these answer choices are incorrect.

(Module 18.2, LOS 18.d)

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## Question #24 - 25 of 57

Question ID: 1675335

Which of the following statements by Plant is *most accurate*?

**A)** Statement 3.

**B)** Statement 2.

**C)** Statement 1.

**Explanation**

Statement 1 is most accurate. Generally, the carry trade is implemented by borrowing in the lower interest rate currencies of developed economies and investing in the higher interest rate currencies of emerging economies. In periods of financial stress, the currencies of the higher risk emerging economies will typically depreciate sharply relative to the currencies of developed economies, and carry trades will generate significant losses.

Statement 2 is incorrect. In the carry trade, the currency trading at a forward premium will have the low interest rate. At the initiation of the carry trade, the carry trader borrows the low interest rate currency. The trader then sells (*not buys*) the currency trading at a forward premium in the spot market.

Statement 3 is incorrect. The carry trade is based on a violation of uncovered interest rate parity (UCIRP). If UCIRP held, then the currency with the higher interest rate will decrease (*not appreciate*) in value by the amount of the initial interest rate differential. In this case, the carry trade would earn a zero return.

(Module 18.2, LOS 18.d)

## Question #25 - 25 of 57

Question ID: 1675336

The July cash flow from the settlement of the INR/NDF is *closest* to:

**A)** INR 1,300,000.

**B)** USD 15,099.

**C)** USD 30,197.

**Explanation**

The cash flow in INR is equal to the notional size of the contract times the future spot rate minus the NDF rate: USD 1,000,000 × (INR 86.10/USD – INR 84.80/USD) = INR 1,300,000.

However, NDFs are settled in a developed market (deliverable) currency. Thus, the INR 1,300,000 is converted to USD at the July spot rate: INR 1,300,000 × USD/INR 86.10 = USD 15,099.

INR 1,300,000 is incorrect because it uses the January spot rate instead of the NDF rate. Doing so would result in a cash flow in INR of USD 1,000,000 × (INR 86.10/USD – INR 83.50/USD) = INR 2,600,000. Converted at the July spot rate, INR 2,600,000 × USD/INR 86.10 = USD 30,197.

(Module 18.6, LOS 18.i)

## Question #26 - 29 of 57

Question ID: 1675323

Based on Exhibit 1 and the weights of U.S. and U.K. investments in Dubois's portfolio, the forecasted portfolio domestic-currency return (measured in EUR) is *closest* to:

**A)** 14.70%.

**B)** 7.89%.

**C)** 14.92%.

**Explanation**

The domestic-currency (DC) return on a portfolio of foreign assets is a function of the return on the foreign assets, the return of the foreign currencies, and the weight of each asset in the portfolio. The DC return on a portfolio of *n* foreign assets can be calculated as:

$$R_{DC} = \sum_{i=1}^{n} w_i \left(1 + R_{FC,i}\right) \times \left(1 + R_{FX,i}\right) - 1$$

where $R_{DC}$ is the DC return on a portfolio of *n* foreign assets; $R_{FC,i}$ is the foreign-currency return of foreign asset *i*; $R_{FX,i}$ is the return of the currency that foreign asset *i* is denominated in; and $w_i$ is the weight of foreign asset *i* in the portfolio (in terms of domestic currency).

When the spot rates are quoted as USD/EUR, the USD (i.e., the foreign currency) is the price currency; therefore, to compute the return on the USD, the spot rates should be converted so that the USD is the base currency.

$$R_{DC} = 35\% \times (1 + 16\%) \left( \frac{\frac{1}{1.2180}}{\frac{1}{1.1171}} \right) + 65\% \times (1 + 14\%) \left( \frac{1.1221}{1.1769} \right) - 1 = 7.89\%$$

The 14.70% answer option is incorrect as the returns on the foreign currencies are omitted, resulting in $R_{DC}$ = 35% × (1 + 16%) + 65% × (1 + 14%) – 1 = 14.70%.

The 14.92% answer option is incorrect as USD/EUR spot rates are not converted to turn the USD into the base currency, resulting in

$$R_{DC} = 35\% \times (1 + 16\%) \left( \frac{1.2180}{1.1171} \right) + 65\% \times (1 + 14\%) \left( \frac{1.1221}{1.1769} \right) - 1 = 14.92\%$$

(Module 18.1, LOS 18.a)

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## Question #27 - 29 of 57

Question ID: 1675324

Based on Exhibit 1, and assuming that GP Group's forecasts are accurate and that Dubois uses 1-year forward contracts to hedge the current value of his foreign assets, to rebalance the hedge at the end of the year, Dubois should *most likely*:

**A)** sell both the USD and GBP forward. ✓

**B)** buy both the USD and GBP forward. ✗

**C)** buy the USD forward and sell the GBP forward. ✗

**Explanation**

To hedge the long USD and GBP exposures, USD and GBP should be sold forward (until the end of the year). By year-end, assuming GP Group's forecasts are accurate, the values of U.S. and U.K. investments will have increased by 16% and 14%, respectively; therefore, to rebalance the hedge, Dubois should sell additional amounts of USD and GBP forward.

Assuming GP Group's forecasts are accurate, because the portfolio's USD and GBP exposures will increase by year-end, rebalancing the hedge requires increasing selling additional amounts of USD and GBP forward.

To rebalance the hedge at the end of the year, Dubois needs to sell both USD and GBP forward.

(Module 18.4, LOS 18.f)

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## Question #28 - 29 of 57

Question ID: 1675325

Based on Exhibit 1 and Dubois's requirements pertaining to hedging the GBP exposure using EUR/GBP currency options, which of the following is the *most suitable* strategy?

**A)** Buy a 50-delta put, write a 25-delta put, and write a 25-delta call. ✗

**B)** Buy a 35-delta call, write a 25-delta call, and write a 25-delta put. ✗

**C)** Buy a 35-delta put, write a 25-delta put, and write a 25-delta call. ✓

**Explanation**

A *seagull spread* is an option strategy that consists of a put spread and a short call. The put spread combines a long out-of-the-money (OTM) or an at-the-money put and a short deeper OTM put to finance of cost of the long put.

Dubois should buy a 1.1650 strike put and write a 1.1550 strike put to be protected if the EUR/GBP trades between 1.1550 and 1.1650. The 1.1650 strike put is OTM and the 1.1550 strike put is deeper OTM because the EUR/GBP is currently at 1.1769; as such, these two puts are most likely 35-delta and 25-delta, respectively. Dubois should also write a 1.1950 strike call (OTM call) that would further reduce the cost of the hedge while forgoing the upside potential if the EUR/GBP rises above 1.1950.

Because Dubois has a long exposure to the GBP, to protect against downside risk, he should buy an EUR/GBP put (rather than a call).

Because the protective put should have a 1.1650 strike and the current EUR/GBP spot rate is 1.1769, the long put is OTM rather than ATM (i.e., 50-delta).

(Module 18.5, LOS 18.g)

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## Question #29 - 29 of 57

Question ID: 1675326

In regard to hedging currency exposures using forwards and options, which of the following is *most accurate*?

**A)** Both statements are correct. ✓

**B)** Only Statement 2 is correct. ✗

**C)** Only Statement 1 is correct. ✗

**Explanation**

*Statement 1 is correct*. While forward contracts do not require an up-front payment, forgoing the upside potential of a favorable currency movement represents an implicit cost. Moreover, hedging currency exposures using forwards may involve a negative roll yield (i.e., the return from the convergence of the forward toward the spot price as the contract nears expiry) if the currency sold (bought) forward is trading at a forward discount (premium).

*Statement 2 is also correct*. Buying options to protect against downside risk requires paying an up-front premium, which may be expensive (especially if the options bought are ATM). As the option moves toward expiry, its time value declines; this is known as *time decay* and works against the holder of a long option position.

(Module 18.5, LOS 18.g)

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## Question #30 - 33 of 57

Question ID: 1675299

Based on **Exhibit 1: One-Year Yields** and **Exhibit 2: One-Year Implied Volatilities**, the *most appropriate* carry trade strategy involves borrowing in:

**A)** EUR and investing in CHF. ✗

**B)** CHF and investing in EUR. ✓

**C)** USD and investing in EUR. ✗

**Explanation**

A carry trade strategy involves borrowing in low-yield currencies (i.e., currencies trading at a forward premium) and investing in high-yield currencies (i.e., currencies trading at a forward discount) to earn the spread between the two yields. This strategy is based on the violation of the uncovered interest rate parity. If the uncovered interest rate parity holds, the high-yielding currency will depreciate on average by a percentage equal to the initial differential between the two yields, and the return of the strategy will be equal to zero. A carry trade strategy is implemented when the volatility of a currency pair is low.

Borrowing in CHF at –0.59% (negative yield) and investing in EUR at 0.49% will generate the highest spread, and the volatility of the CHF/EUR is the lowest.

Borrowing in EUR at 0.49% (the highest yield) and investing in CHF at –0.59% (the lowest yield) will generate a negative spread.

Borrowing in USD and investing in EUR will generate a lower spread than will borrowing in CHF and investing in EUR. Moreover, the volatility of the USD/EUR is higher than the volatility of the CHF/EUR.

(Module 18.2, LOS 18.d)

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## Question #31 - 33 of 57

Question ID: 1675300

Kumar's statement pertaining to volatility-based trading strategies is *most likely* incorrect with regard to:

**A)** the payment of options premiums.

**B)** long volatility.

**C)** short volatility.

**Explanation**

In a volatility-based trading strategy, investors trade volatility risk by using options based on their views regarding future volatility.

Typically, hedgers want to protect their portfolios against an increase in volatility and are long volatility by establishing long options positions and paying the options premiums.

On the other hand, speculators are typically short volatility by writing options and earning (rather than paying) the options premiums. If the short options expire out of the money, which is usually the case (because volatility is relatively low under normal market conditions), the speculators keep the premiums. Ideally, speculators turn from short to long volatility ahead of an increase in volatility; however, timing the market is extremely difficult.

Long options positions should be established to long volatility when anticipating an increase in the foreign exchange rate's volatility.

Traders typically short volatility by establishing short options positions to earn the options premiums when foreign exchange markets are expected to remain stable (i.e., low volatility is expected).

(Module 18.2, LOS 18.d)

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## Question #32 - 33 of 57

Question ID: 1675301

Based on **Exhibit 3: Selected Spot and Forward Rates**, and assuming Graham is a risk-neutral investor, Kumar should *most likely* recommend hedging:

**A)** the ZAR exposure, but not hedging the EUR exposure. ✓

**B)** both the EUR and the ZAR exposures. ✗

**C)** the EUR exposure, but not hedging the ZAR exposure. ✗

**Explanation**

A manager will hedge the currency exposure when the hedge is expected to enhance the portfolio's return. The following table summarizes the unhedged and hedged EUR and ZAR returns.

| | Hedged Return | Forecasted Unhedged Return |
|---|---|---|
| EUR | $\frac{1.2173}{1.2123}$ | $\frac{1.2173}{1.2123} - 1 = 0.41\%$ |
| ZAR | $\left(\frac{1/14.4726}{1/14.1055}\right) - 1 = -2.54\%$ | $\left(\frac{1/14.6889}{1/14.1055}\right) - 1 = -3.97\%$ |

- The EUR is trading at a forward premium of 0.41% against the USD. Hedging the EUR exposure involves selling the EUR at a forward price of 1.2173, which is higher than the current spot price, generating a positive roll yield. However, Kumar expects the EUR to appreciate against the USD to 1.2370 (by 2.04%). Because Graham is a risk-neutral investor and the expected appreciation of the EUR is higher than the expected benefits of hedging (i.e., the roll yield), he would not hedge.
- The ZAR is trading at a forward discount of 2.54% against the USD. Hedging the ZAR exposure involves selling the ZAR at a forward price of $\frac{1}{14.4726}$, which is lower than the current spot price $\frac{1}{14.1055}$, of thus generating a negative roll yield of –2.54%. Moreover, Kumar expects the ZAR to depreciate by 3.97% against the USD. Based on Kumar's expectation, Graham should hedge the ZAR exposure because the cost of hedging (the negative roll yield) is less than the expected depreciation of the ZAR.

Kumar should recommend not hedging the EUR exposure and should recommend hedging the ZAR exposure.

(Module 18.4, LOS 18.f)

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## Question #33 - 33 of 57

Question ID: 1675302

Based on **Exhibit 4: BRL/USD Spot and Forward Rates** and the information on Graham's Brazilian assets, the difference between the hedged return and the unhedged return on the Brazilian assets is *closest* to:

**A)** –3.54%. ✗

**B)** –3.20%. ✗

**C)** –3.74%. ✓

**Explanation**

The difference between the hedged return and the unhedged return is the return on the forward contract (as a percentage of the initial investment in USD).

The initial investment in BRL was 15 million on April 1, 2020, which was equivalent to $\frac{15{,}000{,}000}{5.4317}$ = $2,761,566. The BRL exposure was hedged by selling a BRL 15 million 3-month forward contract.

Loss from the forward contract = $\left(\frac{1}{5.4717} - \frac{1}{5.2728}\right)$ × 15,000,000 = −$103,410.

Thus, return on the forward contract = $\frac{-103{,}410}{2{,}761{,}566}$ = −3.74%.

The return on the forward contract is miscalculated as a percentage of the ending (rather than the beginning) value of the assets in USD. The ending value of the Brazilian assets in USD = $\frac{15{,}390{,}000}{5.2728}$ = $2,918,753 and the resulting return on the forward contract = $\frac{-103{,}410}{2{,}918{,}753}$ = −3.54%.

The loss on the forward contract is miscalculated using the current 3-month forward rate instead of the current spot rate as $\left(\frac{1}{5.4717} - \frac{1}{5.3008}\right)$ × 15,000,000 = −$88,383, resulting in the return on the forward contract = $\frac{-88{,}383}{2{,}761{,}566}$ = −3.20%.

(Module 18.4, LOS 18.f)

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## Question #34 of 57

Question ID: 1675289

The one-year interest rate is 3% in Country A and 6% in Country B. Assuming covered interest rate parity holds:

**A)** Country A's currency will trade at a forward discount. ✗

**B)** the percentage appreciation in Country A's currency will be greater than the percentage depreciation in Country B's currency. ✓

**C)** the carry trade will result in a positive return. ✗

**Explanation**

Assuming covered interest rate parity holds, the currency of the country with the higher interest rate will depreciate (Country B). Country A's currency will appreciate by (1.06 / 1.03) – 1 = 0.0291, or 2.91%. Country B's currency will depreciate by (1.03 / 1.06) – 1 = –0.0283, or 2.83%.

Country A's currency will trade at a *forward premium* because it is the country with the lower interest rate, while Country B's currency will trade at a *forward discount*.

The *carry trade* is based on a violation of *uncovered interest rate parity*, which says the forward exchange rate that is calculated through the covered interest rate parity is an unbiased estimate of the spot exchange rate (therefore, the carry trade would earn a zero return).

(Module 18.2, LOS 18.d)

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## Question #35 of 57

Question ID: 1675330

Which of the following statements regarding managing emerging-market currency exposures is correct?

**A)** Transaction costs for emerging-market currencies are generally similar when comparing exiting trades versus entering trades.

**B)** On the assumption on increasing yields for emerging-market currencies, it will lead to negative roll yield for investors who are selling such currencies forward.

**C)** Value at risk is an appropriate measure of risk for emerging-market currencies.

**Explanation**

Increasing yields will lead to larger forward discounts. The result is negative roll yield for investors who are required to sell such currencies forward.

(Module 18.6, LOS 18.i)

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## Question #36 of 57

Question ID: 1675308

Jane Simms manages a German portfolio denominated in the EUR and decides to use an option collar to reduce her downside risk exposure to the Mexican peso (MXP) while retaining some potential upside. Which of the following strategies will accomplish her objective?

**A)** Buy an OTM put on the MXP and sell an in-the-money call on the MXP.

**B)** Buy an OTM put on the MXP and sell an OTM put on the MXP with a lower strike price.

**C)** Buy an OTM call on the EUR and sell an OTM put on the EUR.

**Explanation**

The collar Simms describes requires buying OTM puts and selling OTM calls on the MXP. However, this is directly equivalent to buying OTM calls on the EUR and selling OTM puts on the EUR. Note that selling an in-the-money call on the MXP removes all portfolio upside and will not meet her objectives.

(Module 18.5, LOS 18.g)

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## Question #37 of 57

Question ID: 1675286

A U.S. investor has some of her wealth invested in a 300,000 euro portfolio managed by a European fund manager. The investor wants to terminate her euro portfolio in nine months but is worried about currency fluctuation; therefore, she wants to hedge the full 300,000 euro value today. The investor should:

**A)** buy euros and buy U.S. dollars under a 9-month forward contract, which is a perfect hedge because it covers the full portfolio value.

**B)** sell U.S. dollars and buy euros under a 9-month forward contract, which is an imperfect hedge because it doesn't hedge potential gains or losses in the portfolio.

**C)** sell euros and buy U.S. dollars under a 9-month forward contract, which is an imperfect hedge because it doesn't hedge potential gains or losses in the portfolio.

**Explanation**

The investor wants to sell her euro portfolio in nine months. A 9-month sell euros/buy U.S. dollars forward contract would achieve this objective. A currency forward contract with an underlying notional value of 300,000 euros would hedge the portfolio value today, but it ignores any potential gains or losses over the next nine months. As a result, the hedge is imperfect, and the investor may have either overhedged or underhedged her portfolio. At the same time, the contract will fix in advance the amount of U.S. dollars to be received in the future.

(Module 18.1, LOS 18.c)

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## Question #38 of 57

Question ID: 1675310

A call option with a strike price of $60 will be available as soon as the stock price of DAG, Inc. (DAG), reaches $55. Which of the following items *best* describes the option on DAG?

**A)** Knock-in option. ✓

**B)** Binary option. ✗

**C)** Knock-out option. ✗

**Explanation**

The call option on DAG is a knock-in option because it only comes into existence if the underlying first reaches some prespecified level (i.e., DAG stock reaches $55).

A knock-out option ceases to exist if the underlying reaches some prespecified level. A binary (or digital) option pays a fixed amount that does not vary with the difference in price between the strike and underlying price.

(Module 18.5, LOS 18.g)

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## Question #39 of 57

Question ID: 1675295

The cost to hedge a long position in the EUR is reduced by:

**A)** forward euro exchange rates that are below the spot exchange rates. ✗

**B)** lower relative interest rates in the euro zone. ✓

**C)** negative roll yield.

**Explanation**

A long position in the euro is hedged by selling the euro forward. Positive roll yield is a reduction in hedging cost. Positive roll yield for the seller of the euro occurs if the forward exchange rate for the euro is above the spot exchange rate. This will occur if euro zone interest rates are relatively low.

(Module 18.4, LOS 18.f)

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## Question #40 of 57

Question ID: 1675328

Luis Gomez manages a Canadian-based investment fund in CAD, which has some U.S.-based investments. Assuming there is a negative correlation between the CAD and corporate profits of the U.S. companies in which it invests, what is the appropriate minimum variance hedge ratio required to reduce the volatility of the returns in CAD?

**A)** Greater than one.

**B)** Less than one.

**C)** One.

**Explanation**

If there is negative correlation between the CAD currency and U.S. corporate profits, then it must mean that the correlation between USD currency and U.S. corporate profits is positive *(note: when we compare USD and CAD currencies to each other, the strengthening of one currency means the weakening of the other currency and vice versa)*.

Therefore, $R_{DC}$ volatility will increase because $R_{FX}$ (USD currency) and $R_{FC}$ (USD corporate profits) are positively correlated. That means there is a greater need to hedge (i.e., hedge ratio must be greater than one) to reduce the volatility of $R_{DC}$.

(Module 18.6, LOS 18.h)

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## Question #41 of 57

Question ID: 1675290

A currency manager is looking to implement currency volatility trades. Which of the following statements about the manager's volatility trades is *most accurate*?

**A)** If cost is a key concern, the manager should implement a long strangle rather than a long straddle. ✓

**B)** If the manager expects a significant rise in volatility, a short straddle will be an appropriate strategy. ✗

**C)** Neither the long straddle nor the short straddle is a delta-neutral strategy. ✗

**Explanation**

A long currency straddle consists of purchasing an at-the-money call and put with the same exercise price, same underlying currency, and same maturity. Therefore, straddles are expensive strategies. Purchasing a strangle is a cheaper alternative, but it requires larger currency movements to be effective. A strangle consists of purchasing an out-of-the-money call and put with the same absolute delta, same underlying currency, and same maturity.

A manager expecting a significant rise in volatility should purchase a long straddle to benefit from either up or down movements in the currency. A short straddle is more appropriate when the manager expects a decrease in volatility.

Straddles and strangles are delta-neutral strategies because the underlying calls and puts have equal but opposite deltas.

(Module 18.2, LOS 18.d)

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## Question #42 of 57

Question ID: 1675294

A U.S.-based investor has purchased a 15,000,000 peso office building in Mexico. He has hedged his investment by selling forward futures at $0.1098/peso. Two months later, the futures exchange rate has fallen to $0.0921/peso. The investor's net change in the futures position is:

**A)** ($265,500). ✗

**B)** $1,647,000. ✗

**C)** $265,500. ✓

**Explanation**

The realized gain on the futures position is:

$$V0(-Ft + F0) = 15{,}000{,}000 \text{ pesos} \times (-\$0.0921/\text{peso} + \$0.1098/\text{peso}) = \$265{,}500.$$

(Module 18.4, LOS 18.f)

## Question #43 of 57

Question ID: 1675284

A French toy manufacturer whose home currency is the euro (EUR) expects to collect payment on a 2 million U.S. dollar (USD) receivable in six months from a U.S. client. The current exchange rate is 0.90 EUR/USD. The market expects the exchange rate to increase to 1.10 EUR/USD, but the manufacturer is worried about any potential currency volatility. Which of the following derivatives contracts would *best* achieve the manufacturer's objectives?

**A)** Buy a 6-month put option to sell USD 2 million at 1.10 EUR/USD.

**B)** Buy a 6-month call option to buy USD 2 million at 1.10 EUR/USD.

**C)** The manufacturer should not hedge because the expected currency movement is favorable to the manufacturer.

**Explanation**

Although the manufacturer expects the USD to strengthen relative to the EUR (i.e., in the future, it will take more EUR to obtain 1 USD), there is concern about potential currency fluctuations and exposure to the risk that the USD does not appreciate as much as expected (or may even depreciate). Buying a put option would allow the manufacturer to benefit from any upside (further appreciation of the USD) while protecting against the downside (depreciation of the USD). The option cost is in the form of payment of an option premium.

The put option is appropriate because the manufacturer is receiving USD, which it needs to then sell for EUR; the call option would only be appropriate if the manufacturer is paying USD on a payable to a U.S. client, for example.

A "do nothing," no-hedge strategy allows the client to benefit from an expected appreciation of the USD but does not provide downside protection against a USD depreciation, which is a risk—especially in more volatile currency markets.

(Module 18.1, LOS 18.b)

## Question #44 of 57

Question ID: 1675287

A Mexican investor with a U.S. dollar portfolio invested in U.S. bonds has a four-month investment time horizon. The investor is concerned about currency volatility. He is contemplating an appropriate and cost-effective currency management strategy and has noted that he is not overly concerned if he misses out on potential currency gains. The investor should:

**A)** engage a currency manager to actively manage his U.S. dollar portfolio.

**B)** leave the U.S. dollar portfolio unhedged.

**C)** fully hedge his U.S. dollar portfolio against currency risk.

**Explanation**

The investor should fully hedge his U.S. dollar portfolio, given that currency volatility can be very high in the short term, especially for an emerging market currency like the Mexican peso. While a fully hedged portfolio would only hedge the portfolio's current market value and leave any portfolio gains or losses unhedged, the investor noted that he is not concerned about missing out on potential currency gains.

A "no hedge" scenario would expose the investor to potentially large currency movements in the short term.

An active currency management strategy is costly, and the costs are likely not justified under a very short time horizon (four months); further, generating currency gains is not one of the investor's objectives.

(Module 18.1, LOS 18.c)

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## Question #45 of 57

Question ID: 1675282

A U.S. investor who holds a £2 million investment wishes to hedge the portfolio against currency risk. The investor should:

**A)** buy £2 million worth of futures for U.S. dollars.

**B)** sell $2 million worth of futures for British pounds.

**C)** sell £2 million worth of futures for U.S. dollars.

**Explanation**

The investor should sell £2 million worth of futures contracts for U.S. dollars. This will offset the existing long position in pound-denominated assets. In so doing, the investor has effectively fixed the exchange rate for pounds into dollars for the duration of the futures contract.

(Module 18.1, LOS 18.b)

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## Question #46 of 57

Question ID: 1675283

Which of the following comments is *most accurate*?

**A)** Currency volatility becomes a more significant issue in global portfolios over a longer time horizon as returns compound. ✗

**B)** A cost/benefit analysis of whether to hedge currency should include all of the following: bid/asked transaction costs, option premiums, and back office and compliance expenses. ✓

**C)** Discretionary currency hedging allows wider deviations from the strategic hedging than active currency management. ✗

**Explanation**

All of the expenses listed should be included in cost/benefit analysis of when and how to hedge currency risk, including items such as back office and other overhead expenses; ultimately, these direct and indirect expenses and costs will affect the client's net return.

The other two answer choices are false. Active management allows the wider deviations. In the long run, currency has less impact on risk as currency tends to mean revert in the long run. A short-run perspective supports currency hedging.

(Module 18.1, LOS 18.b)

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## Question #47 - 52 of 57

Question ID: 1675276

The expected return of the risk-free portion of the portfolio is *closest* to:

**A)** 2.82%. ✓

**B)** 3.74%. ✗

**C)** 6.56%. ✗

**Explanation**

Here are the expected returns measured in the investor's domestic currency (CHF):

EUR asset: (1.02)(0.99) − 1 = +0.98%

ASD asset: (1.025)(1.03) − 1 = +5.58%

The weighted average return is 0.6(0.98%) + 0.4(5.58%) = 2.82%.

(Module 18.1, LOS 18.a)

## Question #48 - 52 of 57

Question ID: 1675277

The standard deviation of the risk-free portion of the portfolio is *closest* to:

**A)** 7.98%.

**B)** 7.36%.

**C)** 54.13%.

**Explanation**

Here are the standard deviations of the risk-free assets measured in the investor's domestic currency:

EUR asset: 7.0%(1.02) = 7.14%

ASD asset: 9.0%(1.025) = 9.23%

The variance of returns of the risk-free portion of the portfolio is:

$0.6^2(7.14^2) + 0.4^2(9.23^2) + 2(0.6)(0.4)(0.70)(7.14)(9.23) = 54.13$

The standard deviation of returns of the risk-free portion of the portfolio is:

$54.13^{½} = 7.36\%$.

(Module 18.1, LOS 18.a)

## Question #49 - 52 of 57

Question ID: 1675278

What is the expected return of the risk-free portion of the portfolio if Archer takes a leveraged position with a 150% positive weight in Australia and a 150% negative weight in the euro zone?

**A)** 9.72%.

**B)** 6.75%.

**C)** 6.90%.

**Explanation**

Here are the expected returns measured in the investor's domestic currency (CHF):

EUR asset: (1.02)(0.99) − 1 = +0.98%

ASD asset: (1.025)(1.03) − 1 = +5.58%

The weighted average return is −1.5(0.98%) + 1.5(5.58%) = 6.90%.

(Module 18.1, LOS 18.a)

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## Question #50 - 52 of 57

Question ID: 1675279

What is the expected standard deviation of returns of the risk-free portion of the portfolio if Archer takes a leveraged position with a 150% positive weight in Australia and a 150% negative weight in the euro zone?

**A)** 98.80%.

**B)** 3.00%.

**C)** 9.94%.

**Explanation**

Here are the standard deviations of the risk-free assets measured in the investor's domestic currency:

EUR asset: 7.0%(1.02) = 7.14%

ASD asset: 9.0%(1.025) = 9.23%

The variance of returns of the risk-free portion of the portfolio is:

$(-1.5)^2(7.14^2) + (1.5)^2(9.23^2) + 2(-1.5)(1.5)(0.70)(7.14)(9.23) = 98.796$

The standard deviation of returns of the risk-free portion of the portfolio is:

$98.796^{½} = 9.94\%$.

(Module 18.1, LOS 18.a)

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## Question #51 - 52 of 57

Question ID: 1675280

If Archer takes a short position in a forward contract in the euro, and assuming the forward exchange rate for the CHF versus the euro is downward sloping, which of the following is *most accurate*?

**A)** She will earn a negative roll return, increasing hedging costs. ✓

**B)** She will earn a negative return by shorting the currency. ✗

**C)** Her position will result in a loss if the euro depreciates more than predicted by interest rate parity. ✗

**Explanation**

Because the forward curve is downward sloping for the CHF exchange rate versus the euro, by taking a short position in the forward contract, she will earn a negative roll return. A positive roll return is earned by taking a long position in a currency with a downward-sloping forward exchange rate, and Archer is taking the opposite position. Conversely, with an upward-sloping forward exchange rate curve, a long position in the forward contract would result in a negative roll return, and a short position would result in a positive roll return. By entering into the forward contract, Archer locks in the forward discount of the euro predicted by interest rate parity. If the euro depreciates more than predicted by interest rate parity, she would lose less by locking in a smaller depreciation of the euro using the forward contract than if she did not hedge with the forward contract. Archer is shorting the forward contract, not the currency itself. If she had shorted the euro currency and it depreciates as expected, then she would earn a profit on the decrease in value of the short position.

(Module 18.1, LOS 18.a)

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## Question #52 - 52 of 57

Question ID: 1675281

Assuming the correlation between the ASD currency and underlying asset is highly positive, which hedge ratio is *most appropriate* for hedging the ASD?

**A)** A hedge ratio <1. ✗

**B)** A hedge ratio >1. ✓

**C)** Cannot be determined with the information given. ✗

**Explanation**

Because the ASD currency and underlying Australian asset are highly positively correlated, a hedging ratio of >1, shorting more than 100% of Australian investment, would result in reduced volatility when measuring the ASD investment in the domestic CHF currency.

(Module 18.1, LOS 18.a)

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## Question #53 of 57

Question ID: 1675311

Mia Gill is a U.S.-based investor and would like to hedge her long position to the Canadian dollar (CAD) using options. Her objective is to minimize the initial option cost by giving up some upside potential, but she does not want to lose any downside protection. Which of the following option strategies is Gill *most likely* to select?

**A)** Seagull spread.

**B)** Put spread.

**C)** Collar.

**Explanation**

A collar could involve buying an out-of-the-money (OTM) put on the CAD and selling an OTM call on the CAD. The OTM put provides downside protection while costing less than an at-the-money (ATM) put. The sale of the OTM call removes some upside potential.

A put spread would involve buying OTM puts on the CAD and then selling puts that are even further OTM. Therefore, the strategy does reduce initial cost, but it also reduces downside protection (compared to buying OTM puts only). There is also no impact on upside potential.

A seagull spread is a put spread combined with selling a call. Compared to the put spread, the seagull spread has an even lower initial cost and the same downside protection (i.e., reduced downside protection), and it reduces some upside potential due to the sale of the call.

(Module 18.5, LOS 18.g)

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## Question #54 - 57 of 57

Question ID: 1675313

Based on **Exhibit 1: Performance Summary of Graham's Portfolio and Selected Spot Exchange Rates**, the domestic-currency return (measured in USD) of Graham's portfolio in 2X20 is *closest* to:

**A)** 22.55%.

**B)** 21.53%.

**C)** 16.06%.

**Explanation**

The domestic-currency (DC) return on a portfolio of foreign assets is a function of the return on the foreign assets (in foreign currency), the return on foreign currencies, and the weights of each asset in the portfolio. The DC return on a portfolio of *n* foreign assets is calculated as:

$$\mathrm{R_{DC}} = \sum_{\mathrm{i=1}}^{\mathrm{n}} \mathrm{w_i} \left(1 + \mathrm{R_{FC,i}}\right) \times (1 + \mathrm{R_{FX,i}}) - 1$$

Where $R_{DC}$ is the DC return on a portfolio of *n* foreign assets; $R_{FC,i}$ is the foreign-currency return on foreign asset *i*; $R_{FX,i}$ is the return on the currency of foreign asset *i*; and $w_i$ is the weight of foreign asset *i* in the portfolio (in terms of domestic currency).

First, compute the weights of the portfolio's EUR- and GBP-denominated assets (based on beginning-of-year values):

**Weights of the EUR- and GBP-Denominated Assets as of December 31, 2X19**

| | Value of the Foreign Asset in the Foreign Currency | Exchange Rate | Value of the Foreign Asset in the Domestic Currency | Weights |
|---|---|---|---|---|
| European (excluding U.K.) equities | EUR 1,510,841 | 1.1202 | USD 1,692,444 | 111.4% |
| U.K. equities* | GBP 131,965 | 1.3113 | USD 173,046 | –11.4% |
| **Total value** | | | **USD 1,519,398** | **100%** |

* Because a short position in U.K. equities is taken, their values and weight in the portfolio are expressed in negative terms.

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Thus, $\mathrm{R_{DC}} = 111.4\% \left(\frac{1{,}637{,}005}{1{,}510{,}841}\right)\left(\frac{1.23}{1.1202}\right) + (-11.4\%)\left(\frac{-111{,}331}{-131{,}965}\right)\left(\frac{1.3613}{1.3113}\right) - 1 = 22.55\%.$

An alternative step-by-step approach is also shown here:

*Step 1*: Calculate the USD starting values.

USD starting value, long EUR position = EUR 1,510,841 × 1.1202 = USD 1,692,444.

USD starting value, short GBP position = GBP 131,965 × 1.3113 = USD 173,045.

Net USD starting value (USD long − USD short) = USD 1,692,444 − USD 173,045 = USD 1,519,399.

*Step 2*: Calculate the USD ending values.

USD ending value EUR long position = EUR 1,637,005 × 1.23 = USD 2,013,516.

USD ending value GBP short position = GBP 111,311 × 1.3613 = USD 151,528.

*Step 3*: Calculate the USD profit on both trades and the overall return.

Profit on long: USD 2,013,516 − USD 1,692,444 = USD 321,072.

Profit on short: USD 173,045 − USD 151,528 = USD 21,517.

Total profit = USD 321,072 + USD 21,517 = USD 342,589.

Overall return = total profit / net amount invested (i.e., long – short):

Overall return = USD 342,589 / USD 1,519,399 = 22.55%.

The 16.06% answer option is incorrect as the value and weight of U.K. equities are erroneously expressed in positive (rather than in negative) terms, resulting in:

$$R_{DC} = \frac{1{,}692{,}444}{1{,}692{,}444 + 173{,}046}\left(\frac{1{,}637{,}005}{1{,}510{,}841}\right)\left(\frac{1.23}{1.1202}\right) + \frac{173{,}046}{1{,}692{,}444 + 173{,}046}\left(\frac{111{,}331}{131{,}965}\right)\left(\frac{1.3613}{1.3113}\right) - 1 = 16.06\%.$$

The 21.53% answer option is incorrect as the end-of-year values (instead of beginning-of-year values) are erroneously used to calculate the weights, resulting in:

$$R_{DC} = \frac{1{,}637{,}005 \times 1.23}{1{,}637{,}005 \times 1.23 - 111{,}331 \times 1.3613}\left(\frac{1{,}637{,}005}{1{,}510{,}841}\right)\left(\frac{1.23}{1.1202}\right) + \frac{-111{,}331 \times 1.3613}{1{,}637{,}005 \times 1.23 - 111{,}331 \times 1.3613}\left(\frac{-111{,}331}{-131{,}965}\right)\left(\frac{1.3613}{1.3113}\right) - 1 = 21.53\%$$

(Module 18.1, LOS 18.a)

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## Question #55 - 57 of 57

Question ID: 1675314

Given Graham's objectives and expectations, and using USD/GBP options, which of the following is the *most appropriate* strategy for hedging his portfolio's exposure to the GBP?

**A)** Only long a 25-delta call and short a 15-delta call.

**B)** Long a 25-delta call, short a 15-delta call, and short a 25-delta put.

**C)** Long a 25-delta put and short a 15-delta put.

**Explanation**

Because Graham has a short position in U.K. equities, he is short the GBP. Therefore, call options can be used to hedge the exposure to the GBP. The strategy in a call spread would meet Graham's objectives because

- the long 25-delta call—an out-of-the-money call—protects the portfolio against moderate unfavorable exchange rates moves;
- the short 15-delta call—a further out-of-the-money call—provides premium income that reduces the initial cost of the strategy; and
- the upside potential of favorable exchange rates moves (i.e., if the USD/GBP falls) is maintained.

Noting that Graham stated that he is prepared to face some downside risk, the strategy involves some downside risk because the currency exposure is unhedged until the USD/GBP reaches the strike of the 25-delta call. Also, noting that Graham stated that he does not expect extreme exchange rates moves, the currency exposure is unhedged beyond the strike price of the 15-delta call.

The choice to long a 25-delta put and short a 15-delta put would be suitable for hedging a long (rather than a short) position in the GBP.

Selling a 25-delta put, limiting the upside potential of favorable exchange rate moves (i.e., if the USD/GBP falls), does not meet Graham's objectives.

(Module 18.5, LOS 18.g)

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## Question #56 - 57 of 57

Question ID: 1675315

Based on the pension fund's IPS, which of the following is the *most appropriate* currency management strategy?

**A)** Currency overlay.

**B)** Active currency management.

**C)** Discretionary hedging.

**Explanation**

Currency management strategies vary from passive hedging to treating the currency as a separate asset class to generate currency alpha. In a discretionary hedging strategy, the manager is given some discretion on how far he or she can deviate from the neutral position (i.e., the benchmark's currency exposures).

Based on the pension fund's IPS, CEAM is only allowed to manage currency exposures while keeping the hedge ratio between 95% and 105%. As such, the appropriate currency management strategy is a discretionary hedging strategy. The primary objective of this strategy is to reduce portfolio currency risk, while giving the manager the ability to have some directional opinion to enhance the portfolio returns. So, discretionary hedging is the correct answer.

A currency overlay strategy refers to outsourcing currency management to a specialized currency manager. This strategy is typically employed when the asset manager does not have significant expertise managing currency exposures. Note that some currency overlay programs may allow the overlay manager to participate in purely speculative trading by treating foreign exchange as an asset class and by taking positions in any currency pair to add value to the portfolio. There is no indication in the text that currency management was outsourced to a specialized currency manager.

The primary objective of active currency management is to manage currency risk to add alpha to the portfolio. Compared to a discretionary hedging strategy, with an active currency management strategy, the manager is allowed to deviate more from the portfolio neutral/benchmark exposures.

(Module 18.1, LOS 18.b)

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## Question #57 - 57 of 57

Question ID: 1675316

Clutch's statement pertaining to hedging the foreign-currency exposure of the pension fund's portfolio is *most likely*:

**A)** incorrect with regard to selling the USD forward.

**B)** incorrect with regard to generating a positive roll yield.

**C)** correct.

**Explanation**

Because the pension fund's portfolio has a long exposure to U.S. equities, it has a long exposure to the USD. Hedging the fund's foreign-currency exposure (i.e., the USD exposure) involves selling the USD against the EUR in a forward contract (i.e., buying EUR forward against USD).

The roll yield is the return generated by the convergence of the forward price toward the spot price as the contract nears expiry. Because the USD/EUR is trading at a forward premium and the EUR is trading at a forward premium against the USD, buying the EUR forward at a price higher than the current spot will generate a negative (rather than a positive) roll yield.

Because the USD/EUR is trading at a forward premium, hedging the foreign-currency exposure using forward contracts will generate a negative roll yield.

Because the pension fund's portfolio has a long exposure to U.S. equities, hedging the USD exposure involves selling the USD against the EUR in a forward contract.

(Module 18.4, LOS 18.f)